# Robotic Integration Technical Report

Generated Date: 1/16/2026

## Source File: part-816e7fcc-53b4-446b-b709-6d6ccdbe322b.step

| **Attribute** | **Value** |
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| **Public URL** | test |
| **COG Offset (mm)** | 67.56 |
| **Uploaded At** | 1/16/2026, 4:20:48 PM |
| **File Path** | c8d704b6-ef89-43d7-ab5a-a857ca66393d/part-816e7fcc-53b4-446b-b709-6d6ccdbe322b.step |

## 1. Physics & Geometry Analysis

| Property | Specification |
|---|---|
| Volume (mm)3 | 345862.83 |
| Material Used | steel |
| Center Of Mass (mm) > X | 31.09 |
| Center Of Mass (mm) > Y | 50 |
| Center Of Mass (mm) > Z | 33.13 |
| Estimated Mass (kg) | 2.715 |
| Inertia Tensor (kg) M2 > Ixx | 0.004755 |
| Inertia Tensor (kg) M2 > Ixy | 0 |
| Inertia Tensor (kg) M2 > Ixz | 0.001324 |
| Inertia Tensor (kg) M2 > Iyy | 0.004777 |
| Inertia Tensor (kg) M2 > Iyz | 0 |
| Inertia Tensor (kg) M2 > Izz | 0.00472 |
| Bbox Max | 100.11, 100.11, 100.11 |
| Bbox Min | -0.11, -0.11, -0.11 |
| Dimensions (mm) > Width Y | 100.22 |
| Dimensions (mm) > Height Z | 100.22 |
| Dimensions (mm) > Length X | 100.22 |
| Part Diagonal (mm) | 173.59 |
| Heuristic Clearance (mm) | 260.38 |

## 2. Picking Analysis & Suction Strategy

Complexity Score: Low | Valid Suction Faces: 5

| Face Index | Area (mm²) | Normal Vector (x,y,z) |
|---|---|---|
| Face 1 | 8000.00 | [1, 0, 0] |
| Face 2 | 8000.00 | [1, 0, 0] |
| Face 3 | 7293.14 | [0, 0, 1] |
| Face 4 | 7293.14 | [0, 0, 1] |
| Face 5 | 2000.00 | [1, 0, 0] |

## 3. Fanuc Controller Manifest

| Manifest Item | Requirement / Standard |
|---|---|
| Fieldbus Options | R639 - PROFINET I/O Device (Siemens Standard), R510 - EtherNet/IP Adapter (Rockwell Standard), J758 - DeviceNet (Legacy Support) |
| Software Packages | R632 - PMC (Internal PLC Logic), R663 - iRVision 2D Guide (Vision Tending), R739 - Intelligent Interference Check |
| Controller Platform | R-30iB Plus (Standard) |
| Integration Handshake | Deterministic 8-byte Discrete I/O map suggested |
| Safety Type | heuristic |
| Required Options > J535 | Collision Guard - Required for Tooling Protection |
| Required Options > J744 | Highly Recommended |
| Required Options > R619 | Safe I/O Unit - Required for Hardware Interlocking |
| Safety Standards | ISO 10218-1:2011, ISO/TS 15066 (Collaborative), ANSI/RIA R15.06 |
| Compliance Rating | Category 3 PLd |
| Stop Time Estimation Ms | 363.6 |
| Heuristic Separation Distance (mm) | 709.72 |
| Maintenance Mode | Deterministic Cycle Counting |
| Battery Replacement | Annual (4x 1.5V D-Size Alkaline) |
| Recommended Lubricant | Fanuc Vigogrease RE0 |
| Grease Replacement Hrs | 11520 |
| Reliability Derating (%) | 95 |

## 4. Integrator Recommendation: Primary Solution

| Robot Specification | Value |
|---|---|
| Model | CRX-10iA |
| Series | CRX |
| Reach (mm) | 1249 |
| Ip Rating | IP54 |
| Cycle Data > Feasible | true |
| Cycle Data > Margin (s) | 33 |
| Cycle Data > Estimated (s) | 12 |
| Payload (kg) | 10 |
| Safety Data > Status | VALID |
| Safety Data > Severity | NORMAL |
| Safety Data > Standard | ISO 13855 |
| Safety Data > Safety Type | heuristic |
| Safety Data > Safety Dist (mm) | 709.72 |
| Payload Utilization | 47.6 |

### Engineering, Integration & Safety

| Integration Parameter | Detail |
|---|---|
| Gripper Type | parallel |
| Input Mass (kg) | 2.715 |
| Design Reach (mm) | 600 |
| Risk Assessment | LOW |
| Design Payload (kg) | 5.24 |
| Input Cog Offset (mm) | 67.56 |
| Calculated Torque Nm | 1.92 |
| Gripper Allowance (kg) | 2.04 |
| Target Plc | GENERIC |
| Native Protocols | EtherNet/IP (Default) |
| Required Fanuc Options | None |
| Safety Dist (mm) | 709.72 |
| Iso 10218 Status | VALIDATED |
| Environmental Rating | IP54 |
| Emergency Stop Category | CAT 3 / PLd |
| Lubricant | Fanuc Vigogrease RE0 |
| Duty Cycle Risk | Standard |

| Integration Parameter | Detail |
|---|---|
| Estimated Mtbf Hours | 80000 |
| Grease Interval Hours | 7680 |
| Joint Inertia > J4 | 1.92 |
| Joint Inertia > J5 | 1.54 |
| Joint Inertia > J6 | 0.96 |
| Software Manifest | J744 - Dual Check Safety (DCS)<br>J535 - Collision Guard |

## 5. Alternative Robot Configurations

| Model | Reach | Payload | Util % | Cycle (s) | Safety Dist |
|---|---|---|---|---|---|
| CRX-10iA/L | 1418 mm | 10 kg | 47.6% | 12 | 709.72 mm |
| M-10iD/10L | 1636 mm | 10 kg | 47.6% | 12 | 709.72 mm |
| M-10iD/12 | 1441 mm | 12 kg | 39.7% | 11.61 | 709.72 mm |

## 6. General Engineering Compliance

| | |
|---|---|
| **Technical Notes** | *Standard industrial ambient conditions.* |
| **Required Ip Rating** | *IP54* |
| **Required Lubricant** | *Standard Fanuc Grease* |
| **Environment Context** | *general* |

# Source File: part-b5a7cfd3-b1ea-4124-9659-94c6ad11694c.step

| Attribute | Value |
|---|---|
| **Status** | success |
| **File ID** | 594c6f6b-0381-459d-9a88-feafeec02469 |
| **Public URL** | test |
| **COG Offset (mm)** | 46.93 |
| **Uploaded At** | 1/16/2026, 4:21:22 PM |
| **File Path** | c8d704b6-ef89-43d7-ab5a-a857ca66393d/part-b5a7cfd3-b1ea-4124-9659-94c6ad11694c.step |

## 1. Physics & Geometry Analysis

| Property | Specification |
|---|---|
| Volume (mm)3 | 2834973.21 |
| Material Used | steel |
| Center Of Mass (mm) > X | 0 |
| Center Of Mass (mm) > Y | 0 |
| Center Of Mass (mm) > Z | 46.93 |
| Estimated Mass (kg) | 22.255 |
| Inertia Tensor (kg) M2 > Ixx | 0.096 |
| Inertia Tensor (kg) M2 > Ixy | 0 |
| Inertia Tensor (kg) M2 > Ixz | 0 |
| Inertia Tensor (kg) M2 > Iyy | 0.096 |
| Inertia Tensor (kg) M2 > Iyz | 0 |
| Inertia Tensor (kg) M2 > Izz | 0.116813 |
| Bbox Max | 110.8, 110, 120.8 |
| Bbox Min | -110.8, -110, -15.8 |
| Dimensions (mm) > Width Y | 220 |
| Dimensions (mm) > Height Z | 136.6 |
| Dimensions (mm) > Length X | 221.6 |
| Part Diagonal (mm) | 340.83 |
| Heuristic Clearance (mm) | 511.25 |

## 2. Picking Analysis & Suction Strategy

Complexity Score: Low | Valid Suction Faces: 5

| Face Index | Area (mm²) | Normal Vector (x,y,z) |
|---|---|---|
| Face 1 | 67858.40 | [-1, 0, 0] |
| Face 2 | 37334.69 | [0, 0, 1] |
| Face 3 | 33929.20 | [-1, 0, 0] |

| Face Index | Area (mm²) | Normal Vector (x,y,z) |
|---|---|---|
| Face 4 | 19085.18 | [0, 0, 1] |
| Face 5 | 12566.37 | [0, 0, 1] |

## 3. Controller Manifest

| Manifest Item | Requirement / Standard |
|---|---|
| Status | No Manifest Data Available |

## 4. Integrator Recommendation: Primary Solution

| Robot Specification | Value |
|---|---|
| Model | UR5e |
| Reach (mm) | 850 |
| Torque Nm | 12.5 |
| Cycle Data > Feasible | true |
| Cycle Data > Estimated (s) | 9 |
| Cycle Data > Derating Factor | 2 |
| Payload (kg) | 5 |
| Safety Data > Msg | Power & Force Limiting (PFL) – ISO/TS 15066 |
| Safety Data > Status | VALID |
| Safety Data > Severity | LOW |
| Inertia (kg)m2 | 0.0583 |
| Utilization (%) | 529.6 |
| Why Recommended | Payload margin: -429.6%, Torque margin: 55.2% |

### Engineering, Integration & Safety

| Integration Parameter | Detail |
|---|---|
| Gripper Type | parallel |
| Input Mass (kg) | 22.255 |
| Design Reach (mm) | 500 |
| Risk Assessment | HIGH |
| Design Payload (kg) | 26.48 |
| Input Cog Offset (mm) | 46.93 |
| Calculated Torque Nm | 12.5 |
| Gripper Allowance (kg) | 4.23 |
| Protocols | EtherNet/IP |
| Target Plc | GENERIC |
| Ur Options | None |
| Safety Dist (mm) | - |
| Iso 10218 Status | VALIDATED |
| Environmental Rating | IP54 |
| Emergency Stop Category | Cat. 3 / PLd (PFL) |
| Risk | High |
| Estimated Mtbf Hours | 26250 |
| Joint Inertia > J4 | 12.5 |

| Integration Parameter | Detail |
|---|---|
| Joint Inertia > J5 | 10 |
| Joint Inertia > J6 | 6.25 |
| Software Manifest | PolyScope 5.x / 6.x<br>URCap SDK |

## 5. Alternative Robot Configurations

| Model | Reach | Payload | Util % | Cycle (s) | Safety Dist |
|---|---|---|---|---|---|
| UR5 (CB3) | 850 mm | 5 kg | -% | 9 | - mm |
| UR10 (CB3) | 1300 mm | 10 kg | -% | 12.6 | - mm |
| UR10e | 1300 mm | 12.5 kg | -% | 12.6 | - mm |

## 6. General Engineering Compliance

| | |
|---|---|
| **Ip Arm** | *IP54 (e-Series) / IP65 (UR20/UR30/UR Series)* |
| **Ip Control Box** | *IP44* |
| **Technical Notes** | *Standard collaborative operation.* |
| **Required Lubricant** | *Sealed (Maintenance Free)* |
| **Environment Context** | *general* |
| **Cleanroom Certification** | *Not specified* |

# Source File: part-fd7e6357-3004-41ce-8c49-88bbb504f7ec.step

| Attribute | Value |
|---|---|
| **Status** | success |
| **File ID** | 49f771ae-3ea5-4e25-b698-e74736946b08 |
| **Public URL** | test |
| **COG Offset (mm)** | 46.93 |
| **Uploaded At** | 1/16/2026, 9:01:37 PM |
| **File Path** | c8d704b6-ef89-43d7-ab5a-a857ca66393d/part-fd7e6357-3004-41ce-8c49-88bbb504f7ec.step |

## 1. Physics & Geometry Analysis

| Property | Specification |
|---|---|
| Volume (mm)3 | 2834973.21 |
| Material Used | steel |
| Center Of Mass (mm) > X | 0 |
| Center Of Mass (mm) > Y | 0 |
| Center Of Mass (mm) > Z | 46.93 |
| Estimated Mass (kg) | 22.255 |
| Inertia Tensor (kg) M2 > Ixx | 0.096 |
| Inertia Tensor (kg) M2 > Ixy | 0 |
| Inertia Tensor (kg) M2 > Ixz | 0 |
| Inertia Tensor (kg) M2 > Iyy | 0.096 |
| Inertia Tensor (kg) M2 > Iyz | 0 |
| Inertia Tensor (kg) M2 > Izz | 0.116813 |
| Bbox Max | 110.8, 110, 120.8 |
| Bbox Min | -110.8, -110, -15.8 |
| Dimensions (mm) > Width Y | 220 |
| Dimensions (mm) > Height Z | 136.6 |
| Dimensions (mm) > Length X | 221.6 |
| Part Diagonal (mm) | 340.83 |
| Heuristic Clearance (mm) | 511.25 |

## 2. Picking Analysis & Suction Strategy

Complexity Score: Low | Valid Suction Faces: 5

| Face Index | Area (mm²) | Normal Vector (x,y,z) |
|---|---|---|
| Face 1 | 67858.40 | [-1, 0, 0] |
| Face 2 | 37334.69 | [0, 0, 1] |
| Face 3 | 33929.20 | [-1, 0, 0] |

| Face Index | Area ($mm^2$) | Normal Vector (x,y,z) |
|---|---|---|
| Face 4 | 19085.18 | [0, 0, 1] |
| Face 5 | 12566.37 | [0, 0, 1] |

## 3. Controller Manifest

| Manifest Item | Requirement / Standard |
|---|---|
| Status | No Manifest Data Available |

## 4. Integrator Recommendation: Primary Solution

| Robot Specification | Value |
|---|---|
| Model | UR30 |
| Series | UR Series |
| Reach (mm) | 1300 |
| Ip Rating | IP65 |
| Cycle Data > Feasible | true |
| Cycle Data > Margin (s) | 35.53 |
| Cycle Data > Estimated (s) | 9.47 |
| Payload (kg) | 30 |
| Safety Data > Status | VALID |
| Safety Data > Severity | LOW |
| Safety Data > Standard | ISO/TS 15066 |
| Safety Data > Safety Type | heuristic |
| Safety Data > Safety Dist (mm) | - |
| Payload Utilization | 94 |

### Engineering, Integration & Safety

| Integration Parameter | Detail |
|---|---|
| Gripper Type | parallel |
| Input Mass (kg) | 22.255 |
| Design Reach (mm) | 694.2 |
| Risk Assessment | HIGH |
| Design Payload (kg) | 28.21 |
| Input Cog Offset (mm) | 46.93 |
| Calculated Torque Nm | 13.36 |
| Gripper Allowance (kg) | 5.95 |
| Target Plc | GENERIC |
| Native Protocols | EtherNet/IP |
| Required Ur Options | None |
| Safety Dist (mm) | - |
| Iso 10218 Status | VALIDATED |
| Environmental Rating | IP65 |
| Emergency Stop Category | Cat. 3 / PLd (PFL) |
| Lubricant | Sealed (No Grease Required) |
| Duty Cycle Risk | High |

| Integration Parameter | Detail |
|---|---|
| Estimated Mtbf Hours | 45500 |
| Grease Interval Hours | N/A |
| Joint Inertia > J4 | 13.36 |
| Joint Inertia > J5 | 10.69 |
| Joint Inertia > J6 | 6.68 |
| Software Manifest | PolyScope 5.x / 6.x<br>URCap SDK |

## 5. Alternative Robot Configurations

| Model | Reach | Payload | Util % | Cycle (s) | Safety Dist |
|---|---|---|---|---|---|
| UR20 | 1750 mm | 20 kg | 141.1% | 15.23 | - mm |
| UR16e | 900 mm | 16 kg | 176.3% | 9.05 | - mm |
| UR10e | 1300 mm | 12.5 kg | 225.7% | 11.95 | - mm |

## 6. General Engineering Compliance

| | |
|---|---|
| **Ip Arm** | *IP54 (e-Series) / IP65 (UR20/UR30/UR Series)* |
| **Ip Control Box** | *IP44* |
| **Technical Notes** | *Standard collaborative operation.* |
| **Required Lubricant** | *Sealed (Maintenance Free)* |
| **Environment Context** | *general* |
| **Cleanroom Certification** | *Not specified* |